Practice workbook · Kilobytes and kibibytes

Kilobyte vs kibibyte and the other prefixes; converting a stated quantity between decimal and binary multiples.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-16

1. Question 2 · part e · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part e only.

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2 Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

(2)

Denary	Hexadecimal
8	
12	

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

(1)

- ☒ **A** 6^2
- ☒ **B** 6×2
- ☒ **C** $6^2 - 1$
- ☒ **D** 2^6

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

(1)

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

(1)

P 7 2 9 3 7 A 0 5 2 4

1. Question 2 · part e · 2 marks

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Worked-example retry · Attempt part e only.

(d) Binary patterns can be interpreted as signed or unsigned integers.

(i) Convert the denary unsigned integer 60 to 8-bit binary.

(2)

(ii) Here is a binary bit pattern for a signed integer in sign and magnitude format.

1001 0110

Convert the binary bit pattern to denary.

Be sure to include a sign symbol in your answer.

(2)

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1. Question 2 · part e · 2 marks

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Worked-example retry · Attempt part e only.

(iii) Negating a signed integer means changing its sign without changing its value.

The negation of +16 is –16. The negation of –24 is +24.

Here is the binary bit pattern for a signed integer in two's complement format.

1110 0101

Convert the binary pattern to its negation in two's complement.

(2)

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(e) Construct an expression to convert 13 kilobytes to kibibytes.

You **do not** need to do the calculation.

(2)

(Total for Question 2 = 13 marks)

1. Question 2 · part e · 2 marks · Working and answer

Working and answer

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2. Question 4 · part a · 2 marks

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Further practice · Attempt part a only.

4 One function of an operating system is file management.

(a) Both kibibyte and kilobyte can be used as measures of file size.

Compare kibibyte and kilobyte.

(2)

(b) An image is 2322 pixels high and 4128 pixels wide.

The image is stored with a 16-bit colour depth.

The metadata for the image is 975 bytes.

Construct an expression to show how the file size, in megabytes, is calculated.

You do **not** need to do the calculation.

(4)

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2. Question 4 · part a · 2 marks · Working and answer

Working and answer

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3. Question 5 · part b.iv · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.iv only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

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1 .....

.....

2 .....

.....

(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

3. Question 5 · part b.iv · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.iv only.

(ii) Explain **one** benefit of writing the software for the smartwatch in a low-level language.

(2)

(iii) Members of the software development team are computing professionals.

Give **two** ways that members of the team can demonstrate ethical behaviour.

(2)

1 .....

2 .....

(iv) The software for the smartwatch takes up 500,000 bytes of storage.

Identify the value that is equivalent to 500,000 bytes.

(1)

- ☒ **A** 0.5 MB
- ☒ **B** 500 KiB
- ☒ **C** 0.5 MiB
- ☒ **D** 5 MB

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14

P 7 8 6 3 7 A 0 1 4 2 0

3. Question 5 · part b.iv · 1 mark

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Further practice · Attempt part b.iv only.

(c) Wearable technology, such as smartwatches, has become very popular with many groups of people including athletes.

Discuss how the use of wearable technology can be a benefit to health.

(6)

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(Total for Question 5 = 16 marks)

3. Question 5 · part b.iv · 1 mark · Working and answer

Working and answer

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